

Name: _____ Date: _____

Period: _____

Primary Objective**I will be able to...**

1. **LO 4.6.A** — Write the standard form equation of a parabola, ellipse (including circles), or hyperbola given key features; and identify those features from a given equation. **write and interpret standard-form equations for all four conic types**

Vocabulary & Key Concepts

conic section: A curve formed by the intersection of a plane and a double cone: a **parabola**, an **ellipse**, or a **hyperbola**.

vertex (parabola): The point (h, k) in the standard form $y - k = a(x - h)^2$ or $x - h = a(y - k)^2$. It is the **turning** point of the parabola.

center (ellipse / hyperbola): The point (h, k) at the **center** of an ellipse or hyperbola; read from h and k in the standard form.

semi-major / semi-minor axis: For an ellipse: the larger radius is a (**semi-major**) and the smaller is b (**semi-minor**).

asymptote (hyperbola): Lines the hyperbola approaches but never reaches: $y - k = \pm - (x - h)$. Fill in b over a : b/a

Hook

Entry Question: Planetary orbits, telescope mirrors, and navigation systems all use conic sections.

Q1. Name one real-world application of a parabola and one of an ellipse. **Parabola: satellite dish / telescope mirror / bridge arch. Ellipse: planetary orbit / whispering gallery.**

Q2. What do you predict all three conic shapes have in common algebraically? **All can be described by second-degree (quadratic) equations in x and y ; all arise from distance relationships.**

Section 1 — Parabolas (LO 4.6.A, EK 4.6.A.1)**Standard forms:**

Form	Axis	Direction
$y - k = a(x - h)^2$	vertical	$a > 0$: opens up ; $a < 0$: opens down
$x - h = a(y - k)^2$	horizontal	$a > 0$: opens right ; $a < 0$: opens left

Worked Example: $y + 3 = 2(x - 1)^2$

Rewrite: $y - (-3) = 2(x - 1)^2$

Vertex: $(1, -3)$ $a = 2$, so opens: **up**

Student Practice: $y - 5 = -(x + 2)^2$
Vertex: $(-2, 5)$ $a = -1$, so opens: **down**

Teacher Note

For the practice problem, students must rewrite $(x + 2)^2$ as $(x - (-2))^2$ to read the vertex correctly. Cold-call: “What is the value of h here?”

Section 2 — Ellipses and Circles (LO 4.6.A, EK 4.6.A.2)

Standard form of an ellipse:

$$\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1$$

Worked Example: $\frac{(x + 1)^2}{16} + \frac{(y - 3)^2}{9} = 1$

Center: $(-1, 3)$ $a^2 = 16 \Rightarrow a = 4$ (horizontal) $b^2 = 9 \Rightarrow b = 3$ (vertical)

Circle (special case): When $a = b = r$, the equation becomes

$$(x - h)^2 + (y - k)^2 = r^2$$

Student Practice: $\frac{(x - 3)^2}{25} + \frac{(y + 1)^2}{4} = 1$

Center: $(3, -1)$ $a = 5$ $b = 2$

Section 3 — Hyperbolas (LO 4.6.A, EK 4.6.A.3)

Standard forms:

Form

Opens

$$\frac{(x - h)^2}{a^2} - \frac{(y - k)^2}{b^2} = 1$$

left and right (x -term positive)

$$\frac{(y - k)^2}{b^2} - \frac{(x - h)^2}{a^2} = 1$$

up and down (y -term positive)

Asymptotes: $y - k = \pm \frac{b}{a}(x - h)$

Worked Example: $\frac{(x - 2)^2}{9} - \frac{(y + 1)^2}{4} = 1$

Center: $(2, -1)$ $a = 3$ $b = 2$ Opens: **left and right**

Asymptotes: $y + 1 = \pm \frac{2}{3}(x - 2)$

Student Practice: $\frac{(y + 3)^2}{16} - \frac{(x - 1)^2}{9} = 1$

Center: $(1, -3)$ $a = 4$ $b = 3$ Opens: **up and down**

Asymptotes: $y + 3 = \pm \frac{4}{3}(x - 1)$

Teacher Note

For the practice hyperbola, $b^2 = 16$ gives $b = 4$ and $a^2 = 9$ gives $a = 3$. The y -term is positive, so the hyperbola opens up and down. The asymptote slope is $b/a = 4/3$ (not a/b). This is the most common error — clarify by writing the formula $y - k = \pm(b/a)(x - h)$ with the denominator labeled.